MAT 1302 - COMPLEX NUMBER EXERCISES
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1. Is $-2/3$ a complex number?

2. Suppose

$$z = 4 - 2i, \quad w = -2 + 3i.$$

Compute the following:

(a) $\bar{z}$, $\bar{w}$
(b) $|z|$, $|w|$
(c) $1/z$, $1/w$
(d) $z + w$
(e) zw
(f) z/w
(g) w/z

3. Write

$$\frac{1 - 2i}{-3 + 5i}$$

in the form $a + bi$.

4. Write

$$\frac{2 + i}{1 - 3i}$$

in the form $a + bi$.